> 415(i). PD estimates for borrowers that are highly leveraged or for borrowers whose assets are predominantly traded assets must reflect the performance of the underlying assets based on periods of stressed volatilities.

4. *Central counterparties*

113. The Committee acknowledges the ongoing work of the Committee on Payment and Settlement Systems (CPSS) and the International Organization of Securities Commissions (IOSCO) to review the 2004 *CPSS-IOSCO Recommendations for Central Counterparties*. Subject to the completion of the revision of the CPSS-IOSCO standards, which cover, among other things, the risk management of a CCP, the Committee will apply a regulatory capital treatment for exposures to CCPs based in part on the compliance of the CCP with the enhanced CPSS-IOSCO standards. The Committee separately will issue for public consultation a set of rules relating to the capitalisation of bank exposures to central counterparties (CCPs). This set of standards will be finalised during 2011, once such consultation and an impact study are complete and after CPSS-IOSCO has completed the update of its standards applicable to CCPs. The Committee intends for these standards to come into effect at the same time as other counterparty credit risk reforms.

5. *Enhanced counterparty credit risk management requirements*

114. Paragraph 36 of Annex 4 will be revised as follows to increase the robustness of banks' own estimates of alpha.

> 36. To this end, banks must ensure that the numerator and denominator of alpha are computed in a consistent fashion with respect to the modelling methodology, parameter specifications and portfolio composition. The approach used must be based on the bank's internal economic capital approach, be well-documented and be subject to independent validation. In addition, banks must review their estimates on at least a quarterly basis, and more frequently when the composition of the portfolio varies over time. Banks must assess the model risk and supervisors should be alert to the significant variation in estimates of alpha that arises from the possibility for mis-specification in the models used for the numerator, especially where convexity is present.

Stress testing

115. The qualitative requirements set forth in Annex 4 for stress testing that banks must perform when using the internal model method have been expanded and made more explicit. More specifically, the existing paragraph 56, Annex 4, of the Basel II text will be replaced with the following:

> 56. Banks must have a comprehensive stress testing program for counterparty credit risk. The stress testing program must include the following elements:
>
> - Banks must ensure complete trade capture and exposure aggregation across all forms of counterparty credit risk (not just OTC derivatives) at the counterparty-specific level in a sufficient time frame to conduct regular stress testing.
> - For all counterparties, banks should produce, at least monthly, exposure stress testing of principal market risk factors (eg interest rates, FX, equities, credit spreads, and commodity prices) in order to proactively identify, and when necessary, reduce outsized concentrations to specific directional sensitivities.

- Banks should apply multifactor stress testing scenarios and assess material non-directional risks (ie yield curve exposure, basis risks, etc) at least quarterly. Multiple-factor stress tests should, at a minimum, aim to address scenarios in which a) severe economic or market events have occurred; b) broad market liquidity has decreased significantly; and c) the market impact of liquidating positions of a large financial intermediary. These stress tests may be part of bank-wide stress testing.
- Stressed market movements have an impact not only on counterparty exposures, but also on the credit quality of counterparties. At least quarterly, banks should conduct stress testing applying stressed conditions to the joint movement of exposures and counterparty creditworthiness.
- Exposure stress testing (including single factor, multifactor and material non-directional risks) and joint stressing of exposure and creditworthiness should be performed at the counterparty-specific, counterparty group (eg industry and region), and aggregate bank-wide CCR levels.
- Stress tests results should be integrated into regular reporting to senior management. The analysis should capture the largest counterparty-level impacts across the portfolio, material concentrations within segments of the portfolio (within the same industry or region), and relevant portfolio and counterparty specific trends.
- The severity of factor shocks should be consistent with the purpose of the stress test. When evaluating solvency under stress, factor shocks should be severe enough to capture historical extreme market environments and/or extreme but plausible stressed market conditions. The impact of such shocks on capital resources should be evaluated, as well as the impact on capital requirements and earnings. For the purpose of day-to-day portfolio monitoring, hedging, and management of concentrations, banks should also consider scenarios of lesser severity and higher probability.
- Banks should consider reverse stress tests to identify extreme, but plausible, scenarios that could result in significant adverse outcomes.
- Senior management must take a lead role in the integration of stress testing into the risk management framework and risk culture of the bank and ensure that the results are meaningful and proactively used to manage counterparty credit risk. At a minimum, the results of stress testing for significant exposures should be compared to guidelines that express the bank's risk appetite and elevated for discussion and action when excessive or concentrated risks are present.

Model validation and backtesting

116. On model validation, the following paragraph (currently in paragraph 42) will be moved after paragraph 40 of Annex 4:

40bis. An EPE model must also include transaction-specific information in order to capture the effects of margining. It must take into account both the current amount of margin and margin that would be passed between counterparties in the future. Such a model must account for the nature of margin agreements (unilateral or bilateral), the frequency of margin calls, the margin period of risk, the thresholds of unmargined exposure the bank is willing to accept, and the minimum transfer amount. Such a model must either model the mark-to-market change in the value of collateral posted or apply this Framework's rules for collateral.